# The Cost of Stepping Back: Children, Financial Facilitation and Wage Outcomes

## Introduction

When a working woman decides to become a mother with young kids, it's a common phenomenon in the labor market for her to be hit with the "motherhood wage penalty**".** This paper strives to determine the dominant mechanism between two competing economic theories to explain this phenomenon.

The first, grounded in Mincer and Ofek's (1982) theory of **human capital depreciation**, predicts that career interruptions caused by young children cause a wage loss by eroding the returns to experience over time. The second, based on Lundberg's (1940) **Added Worker Effect**, predicts that a husband's higher earnings allow a woman to step back. Not by penalizing her skills, but by lowering her reservation wage and enabling her to prioritize time with young kids over a higher wage

These are 2 fundamentally different mechanisms directly effecting wage. One operates at the labor market level based on how employers value a woman's accumulated experience when children are present. The other operates at the household level based on how a family's financials enable the women to choose between more time with her kids and the highest wage more freely (it's no longer a do or die to get money on the table to sustain the family).

**Why it Matters?**

Distinguishing between them matters because they imply entirely different policy responses. If depreciation dominates, the solution could be promoting better childcare access and career continuity for women. If household facilitation dominates, the solution could be to promote household income structure and reducing gender pay equity upstream.

Furthermore, the gender wage gap remains one of the most persistent empirical regularities in labor economics. Based on "*A Grand Gender Convergence" by Claudia Goldin (2014)* a large share of the wage gap between men and women emerges post childbirth. The gap between childless young men and women was much smaller. Hence, understanding why motherhood suppresses wages is not merely an academic exercise but an important step we must take as a society to facilitate the very act that allows us all to continue as a species.

This paper tests these two mechanisms using the **Mroz (1987) Female Labor Supply dataset**, a well-known dataset in labor economics known precisely for its sample selection problem inherent in wage studies of women.

## Data Description

The data comes from the **Mroz (1987) Female Labor Supply dataset** sourced from William Greene's *Econometric Analysis* (7th edition) website at New York University. The dataset contains **753 observations** of married women from the Panel Study of Income Dynamics (PSID) for the year 1975. It contains 19 variables capturing individual stats (age, education, experience), household stats (number of children, husband's wages), and labor market outcomes (hours worked, wage).

### Cleaning and Sample Restrictions

The dataset required minimal surface level preprocessing, as it is a well-structured and widely used benchmark in labor economics. However, several key transformations were implemented to prepare the data for analysis.

An obvious feature of this dataset is that wages are only observed for women who chose to work. Of 753 women in the sample, 428 (56.8%) participated in the labor market and have non-zero wages. The remaining 325 (43.2%) have wages coded as zero, not because their true wage is zero, but because they did not enter the labor market and their wage is therefore unobservable. This is a case of Missing Not at Random (MNAR) as the likelihood of observing a wage is directly tied to the individual's labor supply decision, which depend on unobservable factors like preferences, constraints, or labor market opportunities as well. Because we can't contextually process these zeroes, we restrict the sample size to the 428 working women. Consequently, all estimated relationships should be interpreted as conditional on labor force participation, rather than representative of the broader population of women.

Upon inspection of the raw wages, it was observed that raw wages are heavily right-skewed (min: $0.13, max: $25.00, mean: $4.18, with high kurtosis). Hence, another step in cleaning the data was applying a logarithmic transformation on wages to address right-skewness and stabilize variance. This log transformation produces a distribution that is approximately normal. Furthermore, the log-linear earnings equation is the standard specification in human capital economics since Mincer (1974). Using it allows direct comparability with existing literature.

### Summary Statistics Table

| Variable | Treated Mean (SD) | Control Mean (SD) | Diff. |
|---|---|---|---|
| Log Hourly Wage | 1.138 (0.901) | 1.198 (0.696) | -0.059 |
| Hourly Wage ($) | 4.449 (4.525) | 4.139 (3.107) | 0.309 |
| Years of Education | 13.377 (2.289) | 12.557 (2.270) | 0.820** |
| Years of Experience | 8.509 (5.434) | 13.677 (8.165) | -5.168*** |
| Age | 34.925 (4.879) | 42.968 (7.531) | -8.043*** |
| Husband's Wage ($) | 7.105 (3.488) | 7.243 (3.587) | -0.139 |
| No. Children Under 6 | 1.132 (0.342) | 0.000 (0.000) | 1.132*** |
| No. Children 6–18 | 1.736 (1.361) | 1.296 (1.302) | 0.440** |
| N | 53 | 375 | |

Treated = Women with Young Children (KL6 ≥ 1) | Control = No Young Children

Note: * = p <0.10, ** = p<0.05, *** = p<0.01 (two-sample t-test, unequal variances)

**Key Variables**

**Dependent Variable (Y)** [log (WW)]: the natural logarithm of the wife's hourly wage.

**D1: Cost of Stepping Back** [KL6 × AX]: the interaction of the number of children under age 6 and the wife's years of prior labor market experience.

**D2: Household Facilitation:** [KL6 × HW]: the interaction of the number of young children and the husband's hourly wage.

**Controls:** Education (WE), Experience (AX), Age (WA), Young Children (KL6), Older Children (K618), and Husband's Wage (HW) (all included separately to isolate the interaction effects.)

## Methodology

### Causal Strategy

The identification strategy is **Ordinary Least Squares (OLS) with interaction terms**, using **HC1** heteroskedasticity-robust standard errors throughout. This model was appropriate for our test as:

- The research question concerns how specific mechanisms (experience returns and household wealth) mediate the relationship between children and wages
- Interaction terms allow the effect of experience and husband's wage to differ based on the presence of young children.

- OLS with robust standard errors is the appropriate estimator when the goal is linear conditional mean estimation.

**Variables and Model Construction**

The empirical specification is constructed to isolate two competing mechanisms through which young children may affect wages: human capital depreciation and household facilitation. The dependent variable is the natural log of hourly wages, $\log(Wage)$, which allows coefficients to be interpreted as approximate percentage changes and addresses the right-skewed distribution of earnings.

The analysis begins with a baseline Mincer equation to capture the fundamental determinants of wages like education, experience, and age. This model establishes the fundamental human capital relationship and serves as a benchmark.

$$np.log(wage) \sim educ + exp + age$$

To test the proposed mechanisms, young children (KL6) are introduced along with two interaction terms. The first interaction, $KL6 \times Exp$, captures whether the presence of young children alters labor market experience, representing the human capital depreciation channel.

$$np.log(wage) \sim kids_young + exp + kl6_x_exp + educ + age + kids_old$$

The second interaction, $KL6 \times HW$, captures whether a husband's wage enables a reduction in labor supply, representing the household facilitation mechanism. While the Added Worker Effect traditionally concerns labor supply decisions like hours spent working and not wages, if a husband earns sufficiently high wages, the household's financial pressure is reduced. This lowers the wife's reservation wage and thus enables her to accept a lower-wage job that offers greater flexibility rather than maximizing earnings. She is no longer in a "do or die" position to bring money to the table and instead, can prioritize time with her young children

$$np.log(wage) \sim kids_young + husb_wage + kl6_x_hw + educ + age + kids_old$$

The final model is a horse race between models where we test $KL6 \times Exp$ VS $KL6 \times HW$ on Log(wage) while holding the same constants as above.

$$
\begin{aligned}
log(wage_i) = {} & \beta_0 + \beta_1(kids_young) + \beta_2(exp) + \beta_3(husb_wage) \\
& + \beta_4(KL6_i \times Exp) + \beta_5(KL6_i \times HW) + \beta_6(educ) + \beta_7(age) \\
& + \beta_8(kids_old) + \varepsilon_i
\end{aligned}
$$

Where:

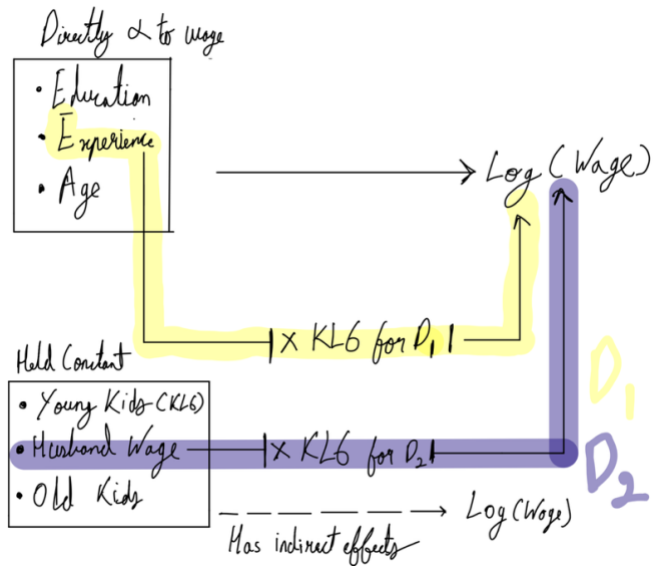
- $\boldsymbol{\beta_4}$ captures D1: does having young children attenuate the returns to experience?
- $\boldsymbol{\beta_5}$ captures D2: does having young children combined with a high-earning husband suppress wages?

The coefficient that is larger in magnitude and more statistically significant is the dominant mechanism

We introduce a full set of controls to isolate these effects. Education is held constant as baseline human capital, experience as accumulated productivity, and husband's wage as a proxy for household income. Age controls for lifecycle effects, and the number of older children (ages 6–18) accounts for differences in childcare intensity across stages of parenting. Holding these variables constant allows the interaction terms to capture variation in wages attributable specifically to the presence of young children through the proposed mechanisms.

**Directed Acyclic Graph**

The causal structure is as follows, education and experience form the **human capital stock** of the wife, which directly determines her wage. Young children (KL6) enter through two pathways: (1) directly via the interaction with her experience (D1), which operates through the labor market's perception of her human capital; and (2) via the interaction with the husband's wage (D2), which operates through the household's financial logistics. Husband's wage also enters independently as a control for associative mating. Age and older children are included as controls for lifecycle effects and caregiving intensity, respectively.




## Results

### Base Model: Mincer Log(wage) only

Before we introduce the aspects of kids, we run a base line model based on the mincer standard human capital theory. Education was the dominant predictor of wages, with each additional year of schooling associated with a 10.9% increase in log wages ($p < 0.001$). Experience adds a statistically significant 1.6% per additional year ($p < 0.01$). Age is not independently significant once experience is controlled, consistent with the theoretical distinction between these variables.

The baseline $R^2$ of 0.149 means education, experience, and age together explain approximately 15% of the variation in log wages among working women (a modest but typical figure for individual-level wage regressions in this era).

### D1: Cost of Stepping back Model

The cost-of-stepping-back mechanism, captured by the interaction Kids young × Exp, is not statistically significant ($\beta = 0.015$, $t = 0.739$, $p = 0.460$). Notably, the sign is *positive* rather than the theoretically predicted negative, suggesting that in this working-women sample, more experienced mothers do not earn less than their childless counterparts. Experience remains independently significant ($\beta = 0.015$, $p < 0.01$), and education continues to dominate ($\beta = 0.110$, $p < 0.001$).

The $R^2$ increases only marginally from the baseline (0.152 vs 0.149), and the AIC *worsens* (879.5 vs 875.4), indicating that adding the D1 interaction does not improve model fit.

**D2: Household Facilitation Model**

The household facilitation mechanism, captured by Kids young × Husband Wage, is also not statistically significant (β = −0.026, t = −0.814, p = 0.416). The sign is now correctly negative which is consistent with the theory that high-earning husbands combined with young children suppress the wife's wages. Husband's wage independently does not reach significance in this model (β = 0.016, p = 0.113).

The $R^2$ falls marginally from our baseline (0.152 vs 0.135) and the AIC for Model 2 is the worst of all four (888.1), suggesting the D2 alone captures or explains less of the variation in log wages than the baseline.

**Horse Race: Both mechanisms together**

With both mechanisms competing simultaneously, the findings are clear and consistent across D1 and D2:

- D1 (KL6 × Exp): β = 0.013, t = 0.647 (not significant)
- D2 (KL6 × HW): β = −0.026, t = −0.839 (not significant)
- Husband's Wage (HW independently): β = 0.023, t = 2.197 (significant at 5%)
- Education: β = 0.101, t = 7.163 (significant at 1%)
- Experience: β = 0.017, t = 3.142 (significant at 1%)

The Horse Race model achieves the best fit ($R^2$ = 0.162, AIC = 878.4) of all the models. The dominant finding is that neither the labor market depreciation mechanism (D1) nor the household facilitation mechanism (D2) is a statistically significant determinant of wages among working mothers in this sample. Instead, the wife's own human capital at labor market entry (principally her education) remains the overwhelming predictor.

Regression Table

| Variable | Baseline | D1 Only | D2 Only | Horse Race |
|---|---|---|---|---|
| Intercept | -0.3469<br>(0.2743) | -0.2167<br>(0.3064) | -0.2434<br>(0.3108) | -0.2165<br>(0.3079) |
| Education (WE) | 0.1093***<br>(0.0134) | 0.1098***<br>(0.0138) | 0.1035***<br>(0.0144) | 0.1014***<br>(0.0142) |
| Experience (AX) | 0.0163***<br>(0.0052) | 0.0145***<br>(0.0052) | — | 0.0165***<br>(0.0052) |
| Age (WA) | -0.0014<br>(0.0055) | -0.0033<br>(0.0060) | 0.0018<br>(0.0052) | -0.0052<br>(0.0062) |
| Young Children (KL6) | — | -0.2174<br>(0.2655) | 0.0947<br>(0.2714) | -0.0152<br>(0.3829) |
| Older Children (K618) | — | -0.0161<br>(0.0294) | -0.0426<br>(0.0282) | -0.0173<br>(0.0292) |

| | | | | |
|---|---|---|---|---|
| Husband's Wage (HW) | — | — | 0.0163<br>(0.0103) | 0.0226**<br>(0.0103) |
| D1: KL6 × AX | — | 0.0147<br>(0.0199) | — | 0.0130<br>(0.0201) |
| D2: KL6 × HW | — | — | -0.0260<br>(0.0320) | -0.0263<br>(0.0314) |
| **R-Squared** | 0.1485 | 0.1524 | 0.1351 | 0.1624 |
| **Adj. R-Squared** | 0.1425 | 0.1403 | 0.1228 | 0.1464 |
| **AIC** | 875.39 | 879.45 | 888.09 | 878.35 |

Note:

- p <0.10, ** = p<0.05, *** = p<0.01
- Robust standard errors (HC1) in parentheses

## Robustness Checks

### Check 1: Outlier Sensitivity

A Tukey Fence analysis identified outliers in key variables:

- log(wage): 20 outliers (4.7%), fence [−0.36, 2.78]
- Experience: 7 outliers (1.6%), fence [−9.50, 34.50]
- Husband's wage: 16 outliers (3.7%), fence [−1.20, 14.86]
- KL6 × Exp: 51 observations (11.9%)
- KL6 × HW: 53 observations (12.4%)

The higher outlier rates for the interaction terms are driven by structural sparsity rather than true extreme values. Because most women have no young children (KL6 = 0), the interaction terms are zero for a large portion of the sample, creating a zero-inflated distribution.

This reduces statistical power but does not bias coefficients. The core results are therefore not driven by extreme observations, but by the limited variation in the treatment condition.

### Check 2: Residual Diagnostics

Normality of Errors:
The Jarque–Bera test is large across all models (JB ≈ 280–305, p ≈ 0), indicating non-normal residuals with left skewness and excess kurtosis. The histogram and Q-Q plot confirm mild tail deviations.

While this violates the normality assumption, it does not affect the unbiasedness of OLS coefficients. However, it may affect standard inference if not corrected.

Heteroscedasticity:
The Breusch–Pagan test confirms the presence of heteroscedasticity:

- LM Statistic: 21.82
- p-value: 0.01

Since $p < 0.05$, we reject homoscedasticity. This justifies the use of HC1 robust standard errors, which ensure valid inference despite non-constant error variance.

Autocorrelation:
The Durbin–Watson statistic is approximately 1.98 across all models, indicating no evidence of serial correlation, as expected in cross-sectional data.

**Check 3: Model Stability Across Specs**

Model stability across specifications serves as an internal robustness test. The key findings remain consistent across all models:

- Education remains the dominant predictor (**$t > 7$** in all models)
- Experience is consistently positive and statistically significant
- Neither interaction term (D1 or D2) is statistically significant in any specification
- Model fit changes minimally ($R^2$ ranges from 0.149 to 0.162)

## Conclusion

**Summary of Findings**

This paper tested 2 competing mechanisms by which young children could impact the wages of working women: (1) Through children eroding experience returns on wage and (2) Through the husband enabling the wife to accept lower wages by providing a high income. The central empirical finding is that neither mechanism is statistically significant in the Mroz (1987) working-women sample. Across 4 OLS models the interaction terms capturing these two pathways failed to achieve significance. Education emerges as the dominant predictor of wages followed by experience, with husband's wage independently significant through the lens of assortative mating.

**Evaluation of Findings**

The ex-ante hypothesis that household facilitation (D2) would prove the stronger and more statistically significant suppressor is not confirmed by the data. Both D1 and D2 are statistically indistinguishable from zero in the base model, and neither clearly dominates the other. The horse race has no clear winner.

**Limitations and Interpretation of Null**

The most compelling explanation for the null result on both interaction terms is sample selection**.** The dataset's 753 women are not just a random cross-section of mothers but are a sample in which 325 women (43.2%) have already made the decision not to work**.** Furthermore, the data set only captures full time employment not part time or commission base income (occupational flexibility var not present). The non-full time working women are precisely the population most likely to have stepped back due to young children and household wealth.

By restricting the analysis to full time working women (N = 428), the regression sample consists of high-attachment, high-ability women who chose to remain employed despite having young children. Among this selected group, children's effects on wages are attenuated, not because the underlying mechanisms are false, but because the most affected women are not measured by the sample.

Another problem this analysis runs into is clearly visible in the bivariate plot: Only a handful (53) women who have a wage have kids. With such few working women having any young children (KL6 > 0) in the sample, the interaction terms KL6 × Exp and KL6 × Husband Wage are estimated from a small subset of observations. This limits statistical power to find measurable differences through these 2 pathways combined with the fact that the sample we are looking at are outliers in the labor market.

## References & Dataset


Dataset: Mroz (1987). "Female Labor Supply Dataset"
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